National University of Computer and Emerging Sciences

Lahore Campus

Data Communication & Networks(EE2007)

Final Exam

Total Time (Hrs.): 3

Total Marks: 50

Total Questions: 5

Date: January 01, 2026

Course Instructor(s)

Mr. Khalid Ijaz

[redacted]

Roll No Section Student Signature

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Attempt all the questions.

CLO 1: Describe Internet architecture and network performance parameters.

Question No.1 **[6 Marks]**

a. **Describe** the sender's transmission rate, in bytes per second, for a network in which two endpoints have a round-trip time of 100 milliseconds. Suppose that the sender transmits five packets per round trip, and each packet has a size of 1500 bytes. **[3 marks]**

b. **Describe** two ways in which the OSI reference model and the TCP/IP reference model are the same. Also, **describe** two ways in which they differ. **[3 marks]**

CLO 2: Explain physical layer concepts in data communication including data rates, line coding schemes, and Shannon's capacity limits.

Question No.2 **[11 Marks]**

a. A data communication system is designed to transmit digital data over a communication channel with a bandwidth of 2 MHz. Due to noise present in the channel, the measured signal-to-noise ratio (SNR) is 63. As a network engineer, you are required to evaluate the performance limits of this channel. Based on the given bandwidth and SNR, **explain** and determine the appropriate maximum achievable bit rate and the corresponding number of signal levels that can be used for reliable data transmission. **[5 marks]**

b. Four digital channels are to be multiplexed using time-division multiplexing (TDM) without synchronization bits. Two channels have a bit rate of 200 kbps, and the other two channels have a bit rate of 150 kbps. Assume that in each TDM frame, 4 bits are taken from each of the first two channels and 3 bits are taken from each of the remaining two channels. **[1.5+1.5+1.5+1.5 marks]**

i. **Explain** how much will be the size of one TDM frame in bits.

ii. **Explain** what will be the appropriate frame rate.

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iii. **Explain** what will be the duration of one frame.
iv. **Explain** and Compute the overall data rate of the multiplexed TDM signal.

CLO 3: Analyze the design principles involved in data link layer.

Question No.3 **[7 Marks]**

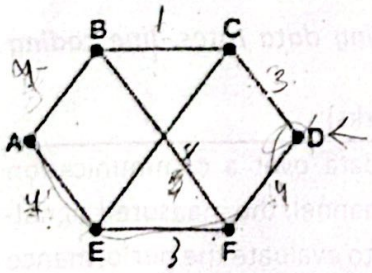
a. In a data communication system, a message represented by the 16-bit sequence 1001 1100 1010 0011 is to be transmitted using the Internet checksum method with 4-bit words. Suppose that the message received by the receiver is 1001 1100 1010 0111. **Analyze** the generation of the checksum code on the transmitter and receiver side and explain whether the transmission is error free or not. **[5 marks]**

b. In a data communication system, a receiver obtains the following data fragment:
0110 0111 1100 1111 0111 1101
The communication protocol uses bit stuffing to prevent accidental flag sequences. **Analyze** the received data and show the original data after performing bit destuffing. **[2 marks]**

CLO 4: Demonstrate the application of network layer protocols.

Question No.4 **[14 Marks]**

a. Consider the following subnet shown below. Distance vector routing is used, and the following link state packets have just come in at router D: from A: (B: 5, E: 4); from B: (A: 4, C: 1, F: 5); from C: (B: 3, D: 4, E: 3); from E: (A: 2, C: 2, F: 2); from F: (B: 1, D: 2, E: 3). The cost of the links from D to C and F are 3 and 4 respectively. **Demonstrate** the final modified routing table of D. Give both the outgoing line to use and the cost. **[8 marks]**



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b. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. **Demonstrate** the first and the last address of the block. Also, **demonstrate** the total number of addresses assigned to that block. **[6 marks]**

CLO 5: Explain network applications and the model of transport layer.

Question No.5 **[12 Marks]**

a. **Explain** the purpose of a TCP segment for each of the following control field values and explain what action the segment indicates in a TCP connection: **[1+1+1+3 marks]**
i. 000000
ii. 000001

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Arbitrary Domain
Attribution Domain

iv. **Explain** and draw the TCP connection termination process between a client and a server. Include all segments exchanged, clearly indicating the use of the FIN and ACK flags, and describe the sequence of events that leads to the graceful closure of the connection.

b. ***Explain*** *the three domains of the domain name space. If a DNS domain name is voyager.fhda.edu, explain how many labels are involved here.* **[6 marks]**

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